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import numpy as np

import matplotlib.pyplot as plt

from sklearn.cluster import KMeans

X = np.array([
    [1, 2], [1, 4], [1, 0],
    [10, 2], [10, 4], [10, 0]
])

kmeans = KMeans(n_clusters=2, random_state=0)

kmeans.fit(X)

labels = kmeans.labels_

centroids = kmeans.cluster_centers_

plt.figure(figsize=(6, 5))

plt.scatter(X[labels == 0][:, 0], X[labels == 0][:, 1],
            s=150, label="Cluster 1")

plt.scatter(X[labels == 1][:, 0], X[labels == 1][:, 1],
            s=150, label="Cluster 2")

plt.scatter(centroids[:, 0], centroids[:, 1],
            marker='X', s=300, label="Centroids")

plt.title("K-Means Clustering (Clear View)")

plt.xlabel("Feature 1")
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plt.ylabel("Feature 2")
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plt.legend()
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plt.grid()
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plt.show()
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